

Comparing Groups

180 minutes · 6 self-check questions

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Choosing the right test for your design, outcome and assumptions — t-tests, ANOVA, chi-square and their non-parametric alternatives — and running them in SPSS and R.

Learning objectives

- Select a test from the design, the outcome type and the number of groups
 - Check the assumptions each test relies on
 - Distinguish paired from independent comparisons
 - Run and interpret t-tests, ANOVA with post-hoc tests, and chi-square
 - Choose a non-parametric alternative when assumptions fail
 - Report a comparison completely: estimate, interval, test statistic and effect size
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Choosing the test

- Three questions settle it: what type is the outcome, how many groups, and are the groups independent or paired?
- Continuous outcome, two independent groups: independent samples t-test
- Continuous outcome, two paired measurements: paired t-test
- Continuous outcome, three or more independent groups: one-way ANOVA
- Categorical outcome, two or more groups: chi-square test of independence
- Small expected counts in a contingency table: Fisher's exact test
- Paired binary outcome: McNemar's test

TEST SELECTION

Outcome	Groups	Parametric	Non-parametric
Continuous	2 independent	Independent t-test	Mann-Whitney U
Continuous	2 paired	Paired t-test	Wilcoxon signed-rank
Continuous	3+ independent	One-way ANOVA	Kruskal-Wallis
Continuous	3+ repeated	Repeated-measures ANOVA	Friedman
Categorical	2+ independent	Chi-square	Fisher's exact
Binary	2 paired	McNemar's test	—

Assumptions, and what to do when they fail

- Independence of observations — the one assumption no test survives losing, and no fix repairs after the fact
- Approximate normality of the outcome within groups; matters most in small samples
- Equal variances for the classic t-test — but Welch's t-test drops this and is a sensible default
- For chi-square, expected counts of at least 5 in most cells
- Judge normality from a histogram, not only a significance test
- When assumptions fail: transform the outcome, use a non-parametric test, or use a bootstrap
- Non-parametric tests compare distributions or ranks, not means, so interpret accordingly

IN PRACTICE

Clustered data — pupils within schools, patients within clinics, repeated visits per person — violate independence. A t-test on such data reports standard errors that are far too small. That needs a mixed model or a cluster-robust approach, not a different t-test.

ANOVA and post-hoc tests

- A significant ANOVA says the groups are not all equal; it does not say which differ
- Running every pairwise t-test afterwards without adjustment inflates the false positive rate
- Post-hoc tests control it: Tukey HSD for all pairwise comparisons, Dunnett when comparing several groups against one control
- Bonferroni is simple and conservative; it divides the significance level by the number of comparisons
- Two-way ANOVA tests two factors and their interaction — an interaction means the effect of one factor depends on the level of the other
- Interpret main effects with care in the presence of a significant interaction

Running it

- R, independent t-test: `t.test(bp ~ group, data = dat)` — Welch by default
- R, paired: `t.test(before, after, paired = TRUE)`
- R, ANOVA: `fit <- aov(bp ~ group, data = dat); summary(fit); TukeyHSD(fit)`
- R, chi-square: `chisq.test(table(dat$group, dat$outcome))`
- SPSS: Analyze > Compare Means > Independent-Samples T Test, or One-Way ANOVA with Post Hoc
- SPSS chi-square lives under Analyze > Descriptive Statistics > Crosstabs > Statistics
- In SPSS always read the Levene test row to decide which t-test line to quote

Reporting a comparison

- Give the group summaries first: mean (SD) or median (IQR) with n per group
- Give the estimated difference with its confidence interval — the interval is the finding
- Give the test statistic, degrees of freedom and exact p-value
- Give an effect size
- Example: "Mean systolic BP was 8.4 mmHg lower in the intervention group (95% CI 3.1 to 13.7; $t(118) = 3.12$, $p = 0.002$; $d = 0.57$)."
- Name the test used and state that its assumptions were checked

Worked examples

Choosing a test in three questions

Four analyses are needed, and the student wants to know which test each requires.

1. Mean BP in intervention vs control, different people: continuous, two groups, independent → independent t-test (Welch).
2. Anxiety score before and after in the same patients: continuous, two measurements, paired → paired t-test.
3. Mean score across four clinics: continuous, four groups, independent → one-way ANOVA, then Tukey.
4. Proportion adhering in two groups: categorical outcome, two groups → chi-square, or Fisher if expected counts are small.
5. Each answer came from three questions: outcome type, number of groups, paired or independent.
6. Only then check assumptions and consider a non-parametric alternative.

THE POINT

Test choice is a short decision tree, not memorisation. Getting the paired versus independent question wrong is the most common error, and it changes both the test and the power.

Pupils in schools are not independent

Test scores from 600 pupils across 20 schools are compared between two teaching methods with an independent t-test. The result is $p = 0.004$.

1. Pupils in the same school share teachers, resources and intake — their scores are correlated.
2. A t-test assumes 600 independent observations. In effect there are closer to 20.
3. Treating correlated observations as independent understates the standard error, so p is far too small.
4. The design effect quantifies it: with an intra-class correlation of 0.15 and 30 pupils per school, variance inflates by about 5.
5. The right analysis is a mixed model with a random intercept for school: `lmer(score ~ method + (1 | school), data = dat)`
6. Refitted properly, the same data give $p = 0.14$ — the original finding was an artefact of ignoring clustering.

THE POINT

Independence is the one assumption no test survives losing, and no transformation repairs. Whenever observations are grouped — schools, clinics, villages, repeated visits — the clustering must be in the model.

Key terms

Welch's t-test

A t-test that does not assume equal variances; a safe default for two independent groups.

Paired test

A comparison of two measurements on the same units, analysing the within-pair differences.

Post-hoc test

A follow-up comparison after a significant ANOVA, adjusted for multiple testing.

Interaction

When the effect of one factor depends on the level of another.

Fisher's exact test

An exact test for a contingency table, used when expected counts are small.

Cohen's d

A standardised difference between two means, expressed in pooled standard deviations.

Clustering

Non-independence arising when observations are grouped within units such as schools or clinics.

Activity

For three research questions from your own study, write down: the outcome type, the number of groups, whether they are paired, and the test you would use. Then name the one assumption most likely to fail in each, and what you would do about it.

Further reading

- **Altman & Bland** — "Parametric v non-parametric methods" — BMJ Statistics Notes, one page, settles most arguments.
- **Delacre, Lakens & Leys (2017)** — "Why Psychologists Should by Default Use Welch's t-test" — The case for dropping the equal-variance assumption.
- **Field** — **Discovering Statistics Using IBM SPSS Statistics** — Step-by-step SPSS with output interpretation.

Self-check questions

1. You measure anxiety in the same 40 patients before and after a programme. The appropriate test is:
 - a. Independent samples t-test
 - b. Paired t-test
 - c. One-way ANOVA
 - d. Chi-square
2. A one-way ANOVA across four groups gives $p = 0.004$. You may conclude:
 - a. All four groups differ from each other
 - b. The groups are not all equal, but a post-hoc test is needed to say which differ
 - c. Group 1 differs from group 4
 - d. The largest and smallest means are significantly different
3. A 2×2 table has expected counts of 3, 4, 12 and 15. The appropriate test is:
 - a. Chi-square
 - b. Fisher's exact test
 - c. Independent t-test
 - d. Kruskal-Wallis
4. You analyse test scores from 600 pupils across 20 schools with an ordinary t-test. The main problem is that:

- a. The sample is too large
 - b. Pupils within a school are not independent, so standard errors will be too small
 - c. Test scores cannot be compared
 - d. A t-test cannot handle 600 observations
5. Levene's test in your SPSS output is significant. This tells you:
- a. The data are not normally distributed
 - b. The group variances differ, so the equal-variances-not-assumed (Welch) row should be read
 - c. The result is not significant
 - d. You must use a chi-square test
6. Which reported result is complete?
- a. The difference was significant ($p < 0.05$).
 - b. Mean BP was 8.4 mmHg lower in the intervention group (95% CI 3.1 to 13.7; $t(118) = 3.12$, $p = 0.002$; $d = 0.57$).
 - c. There was a significant effect of the intervention on blood pressure.
 - d. ANOVA showed $p = 0.002$.
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Answers

1. **b.** The two measurements come from the same people, so they are paired. A paired t-test analyses the within-person differences, which removes between-person variation and is more powerful than treating the groups as independent.
 2. **b.** ANOVA is an omnibus test. It rejects the hypothesis that all means are equal without identifying where the differences lie — that requires an adjusted post-hoc comparison such as Tukey.
 3. **b.** Chi-square relies on an approximation that becomes unreliable when expected counts fall below about 5. Fisher's exact test computes the probability directly and has no such requirement.
 4. **b.** Pupils in the same school share teachers, intake and environment. Ignoring that clustering treats correlated observations as independent, understating standard errors and overstating significance. A mixed model or cluster-robust errors are needed.
 5. **b.** Levene tests homogeneity of variance, not normality. When it is significant, quote the Welch row. Many statisticians now recommend Welch's t-test as the default regardless.
 6. **b.** A complete report gives the estimate, its confidence interval, the test statistic with degrees of freedom, the exact p-value and an effect size. The others tell the reader almost nothing about magnitude or precision.
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